What the council found

Four meditations by Claude Opus, criticized by Claude Fable and sol, 27 October 2026

The frame of the reading of 27 October, three parties against a random-oracle reference, does not survive as a theory. Its single-party notion is a known one, its learning one-way function is empty for the trainers that matter, its filter slogan is false, and its degree is a known exponent. Every critic named the same gap independently. No claim prices a learning attacker, a shadow model or a VAE, in learning units. What survives is a set of exact statements, five corrections to what we have, three rulings for you, and one object that has the shape you asked for. argon's critiques follow tomorrow. Everything below is proven unless labelled otherwise, and "accepted" means accepted by the critic named. Files are in `lab/theory/council/2026-10-27-theory-directions/`.

1. The four topics

Pseudorandomness for learner classes. The notion "a learner class cannot tell the joint law from the product of its marginals" is outcome indistinguishability of the constant predictor. In each class it reduces to a known quantity: total variation, statistical dimension, or the generative exponent. It also equals learning only in log loss, not in square loss. The labels $Y = \langle u, X\rangle \xi$, with ξ an independent sign, are detectable at exponent 2 while no predictor gains in square loss. The positive result, the strongest idea for both critics, is that a Bayes release answers every correlational query exactly as the data do. Hence a square-loss gradient learner pays the same to extract it as to train on the data (accepted by fable, sol asks for integrability hypotheses).

The degree of a channel. The degree of a channel is the generative exponent of Damian, Pillaud-Vivien, Lee and Bruna (DPLB24) with the secret as the label, and the shadow-learning price is their lower bound (accepted by both). The seed cannot supply a graded price. Drawn seeds give one mixture law, and a named seed gives degree 1. A targeted attack on one candidate record has no learning price at all (accepted by both). Every degree above 2 comes from the moments of the data law along a hidden direction. So the frame moves the planted structure into the record law and does not remove it. No mechanism of real deep training was found that cancels low-degree signal, and the meditator's claim that symmetry caps the degree at 2 was rejected by sol with a counterexample. fable's beautiful version is that a degree exists exactly when the hidden index lives on a homogeneous space. The correlation kernel is then zonal, and Hermite and Hahn polynomials are the two cases we have (heuristic in its exact match).

The trainer as a filter. The slogan "what the trainer did not learn stays hidden" is false. A trainer that outputs one bit, or an ε-differentially private bit, hands over a planted parity. A filter must restrict what the trainer does while it trains, its access, its state or its trajectory, never only its output. Under Gaussian noise the attacker's advantage has two terms, a population term and the information the trajectory holds about the sample, and a hashed-table trainer shows the second term is needed (accepted by fa-

ble). fable's verdict is that the composed bound is differential-privacy accounting with Fano's inequality, vacuous for every deployed run.

The learning oracle as a random oracle. Four of the five random-oracle techniques already have instances in the theory, so the reading is mostly language. Programming gives new statements, and they show that an admissible programming changes no price. The sharpest result is that a real seed is a short key. For seeds whose representation laws are separated, about 400 representations identify a 64-bit seed, and every seed-blind SQ lower bound is capped at the number of seeds (accepted by fable, sol notes seeds with identical laws stay hidden).

2. Corrections to what we have

The numbers refer to the reading of 26 October.

1. Item 20 is not a learning one-way function. Against a square-loss gradient trainer the extraction gap vanishes. It exists only against a trainer that reads squared labels, and it is bounded by the information exponent of the label law. The degree-4 link is also a knife edge. Every even link monotone on the half line has generative exponent at most 2, because each level set is a pair $\{\pm a\}$ on which the second Hermite moment is nonzero (fable, checked by the integrator).
2. Item 21 covers SGD only in the worst-rounding model of AKMSS21, in which memorization cannot be stated. With real memorization the statement needs the second term of the noisy filter.
3. Items 12 and 15 are results about SGD. Adam is equivariant only under signed permutations, which have invariants of degree 4 beyond the Gram matrix. On a real implementation the seed-blind statistical and SQ results describe the idealized seed only.
4. "An efficient learner outputs a cheap predictor" holds for circuits and fails for uniform descriptions.
5. The exponent alone is not a unit. At one degree the price is $d^{k/2}$ in a Gaussian frame and d^k in a subset frame, so decision 3 of the definitions reading should not be taken as posed.

3. Three rulings for you

1. Short seeds. The polynomial-time lattice results survive on real implementations only if the implementation's seed generator is pseudorandom for the attacker composed with one training run. Does that assumption count as the cryptographic plug-in you excluded?
2. The programming rule. Read narrowly, a programmed value indistinguishable from an honest sample, it admits canaries, recorded seeds and known noise as measurement devices, and by construction it changes no price. It then can neither admit a defence nor exclude a plug-in. Read broadly, a modification that gives a known law, it admits release noise as a defence. Which did you mean?
3. Canaries. An honest draw measures the average leakage of a typical record, and no published exposure experiment uses such canaries. An out-of-distribution canary measures the worst case, as differential-privacy audits do. Which target do canaries serve?

4. The object with the shape you asked for

Take Gaussian inputs and labels $Y = \langle u, X\rangle \xi$, whose conditional mean is zero. A linear model trained by SGD in square loss has a population drift free of u, so noisy gradient descent releases a model whose

law carries nothing about u. The covariance of its batch updates, however, carries the spike $2uu^\top$ (recomputed by fable). An observer of the updates, such as a federated server, should therefore recover u after about $d(1 + b\varsigma^2)^2$ steps, with b the batch size and ς the injected noise, and fail below that count. That threshold is heuristic, from spiked-covariance scaling, and it is the statement to prove.

The object has a proven part, an open part and an experiment the oracle can run. It prices a learning observer in steps against dimension, batch and noise. It is also exactly what the Bayes-release theorem leaves over, since data and release differ only through the second moment of the labels. It would read the threshold of AKMSS21 as the point where a second-degree channel closes. sol phrases the same programme generally: which parts of the training transition law differ between two experiments, and what those differences cost the attacker.